Lab 3: Parameter Identification (Nonlinearities in Models)

Objectives You model the DC motor with actuator limits and redefined equivalent friction, you will identify the torque needed to overcome the static friction to get the Coulomb friction parameter and find the viscous friction coefficient, you include these nonlinearities in your model, and verify your model with the real system, you estimate motor resistance and motor back-emf constant.

Timeline

This lab spans two weeks. Complete the pre-lab tasks **before the first lab session**. Review all reading materials **during the two weeks but before attending each lab session**. Do not wait until the lab sessions to complete these preparations.

Pre-Lab Tasks

Due: Before the first lab session. Note: You are not required to upload anything to D2L. This pre-lab is for your own practice and preparation only.

Complete the following tasks:

1. Use `polyfit` and `polyval` to fit a quadratic polynomial to the data points: $(x, y) = \{(0, 1), (1, 2), (2, 5), (3, 10)\}$. Plot both the original data points and the fitted curve.
2. Define the transfer function $H(s) = \frac{1}{s+1}$ in MATLAB using the `tf` command.
3. Plot the step response of the above transfer function using the `step` command.
4. Take the following MATLAB course Linearization of Nonlinear Systems, about 30 minutes
   `https://matlabacademy.mathworks.com/details/linearization-of-nonlinear-systems/otmlsllns`

MATLAB Functions to Review:

When visiting the MathWorks documentation pages, please ensure that you review the examples provided. You may also use the features such as "Open in MATLAB Online" or copy the example code into your own installed MATLAB software and practice.

1. Polynomial Fitting and Evaluation

- **Learning Goal:** Understand how to fit a polynomial to given data and evaluate the fitted polynomial.
- **Key MATLAB Functions:**
  - `polyfit`: Fits a polynomial of a given degree to data.
    Documentation: `https://in.mathworks.com/help/matlab/ref/polyfit.html`
  - `polyval`: Evaluates a polynomial at specified points.
    Documentation: `https://www.mathworks.com/help/matlab/ref/polyval.html`

- **Suggested Resource:**
  - MATLAB Curve Fitting Onramp (Length: About 1 hour): `https://matlabacademy.mathworks.com/details/curve-fitting-onramp/orcf`

2. Transfer Function Representation

- **Learning Goal:** Learn to create and manipulate transfer functions.
- **Key MATLAB Function:**
  - `tf`: Creates transfer functions from numerator and denominator coefficients. Documentation: `https://www.mathworks.com/help/control/ref/tf.html`

3. Step Response Analysis

- **Learning Goal:** Understand and visualize the time-domain response of systems to a step input.
- **Key MATLAB Function:**
  - `step`: Computes and plots the step response of dynamic systems. Documentation: `https://www.mathworks.com/help/control/ref/dynamicsystem.step.html`

Required Reading

Complete during the two-week lab period, before attending lab sessions.

From the Reference Book, *Feedback Control of Dynamic Systems* (8th edition), review the following topics:

1. Dynamic Modeling (Mechanical, Electrical, and Electromechanical), Sections: 2.1, 2.2, 2.3
2. Poles and Zeros: Section 3.18
3. Linear System Analysis: Section 3.19

Read the following materials on D2L:

1. Nonlinearities – 11 pages – From Chapter 2, Book: Control System Engineering by Nise, 2nd. Edition (11 pages). This covers Nonlinearities, Linearization of functions and differential equations
2. Mechanical Elements and Friction – 22 pages – From Chapter 3, Book: System Dynamics by Ogata, 3rd Edition. This covers modeling mechanical elements and friction.